ATELIER · AUCTIONS

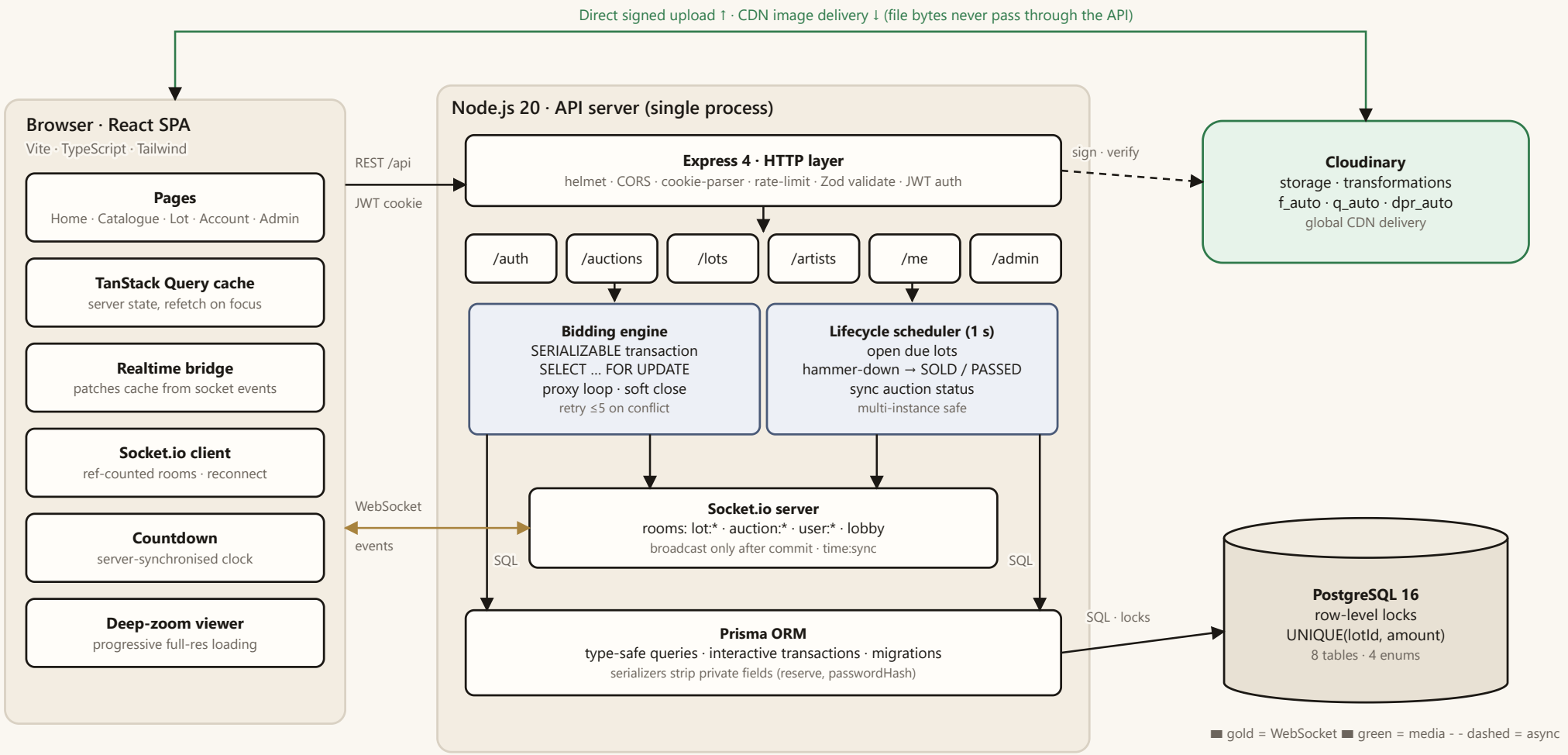
System Architecture & Data Design

A real-time online painting auction platform: live bidding with strict transactional integrity, proxy bids, anti-sniping soft close, server-synchronised countdowns and a Cloudinary image pipeline with deep zoom.

1. High-Level Design (HLD)
2. Entity-Relationship (ER) Diagram
3. Database Schema — identity, catalogue & enums
4. Database Schema — lots, bids & images
5. Flowchart — Bid placement (transactional engine)
6. Lot lifecycle state machine & scheduler flow
7. Sequence — Live bid propagation
8. Sequence — Image upload & clock sync

Backend Node.js 20 · Express 4 · TypeScript · Socket.io 4
Data PostgreSQL 16 · Prisma 5 (SERIALIZABLE tx, row locks)
Frontend React 18 · Vite · Tailwind · TanStack Query 5 · Framer Motion
Auth JWT in HttpOnly cookie · bcrypt · Zod validation · rate limits
Media Cloudinary (signed direct upload, f_auto/q_auto, CDN)
Repo npm workspaces monorepo: */server* · */client*

1 · High-Level Design

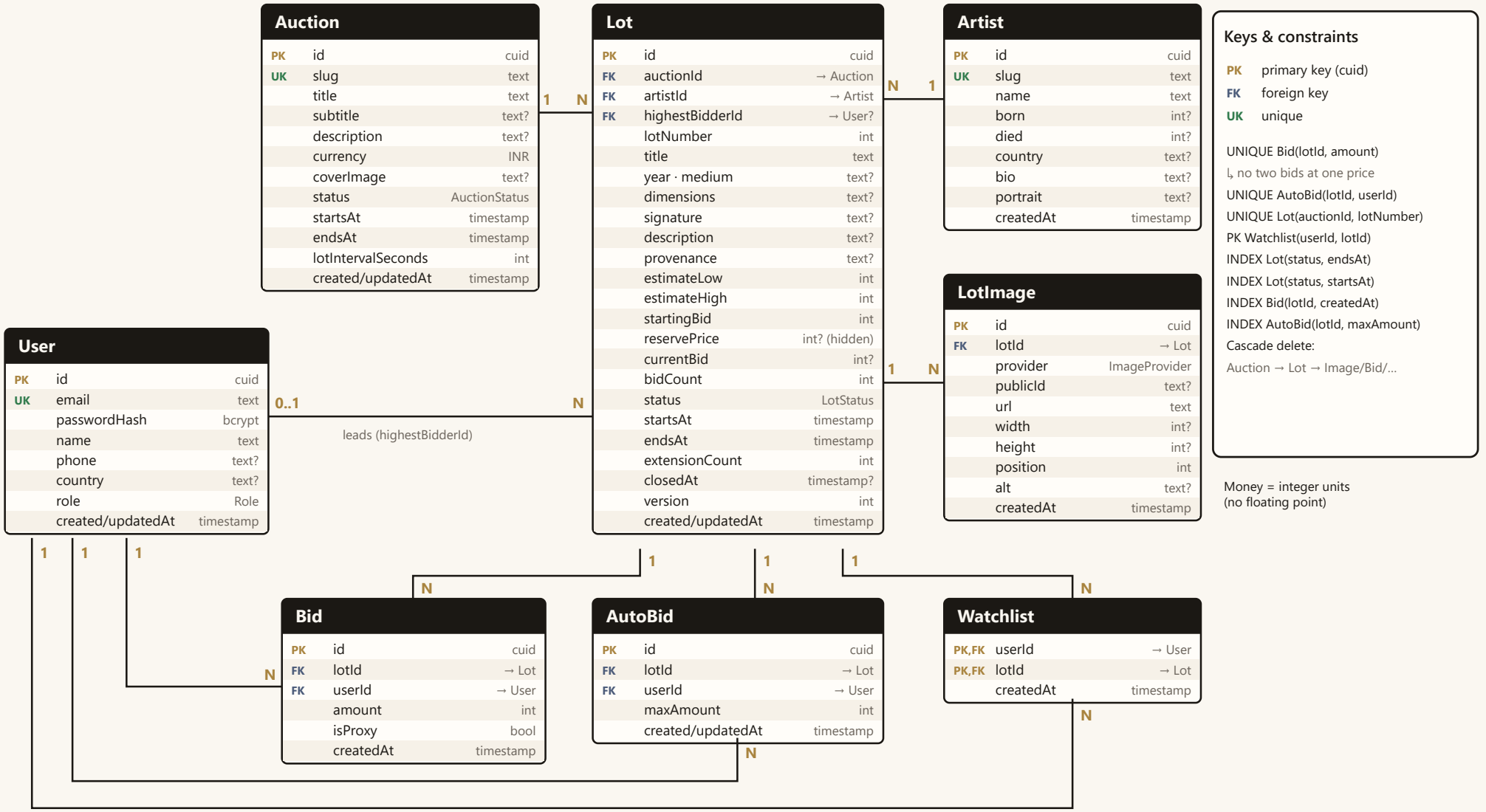


Reads go REST → route → Prisma → serializer (hides reserve price & password hash). **Writes** (bids) run in one SERIALIZABLE transaction holding a row lock on the lot; events are broadcast only *after commit*.

Real time: Socket.io rooms per lot, per auction and per user. One client-side bridge patches every TanStack Query cache entry containing the lot; stale events are dropped by `version`.

Images go browser → Cloudinary directly with a server-issued signature; the API verifies the asset before attaching it. The database is the single source of truth; sockets are only a notification channel.

2 · Entity-Relationship Diagram



3 · Database Schema (1/2)

User

BIDDERS & ADMINS

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
UK email	TEXT	login identity
passwordHash	TEXT	bcrypt, cost 12
name	TEXT	
phone	TEXT?	
country	TEXT?	
role	Role	default BIDDER
createdAt	TIMESTAMP(3)	default now()
updatedAt	TIMESTAMP(3)	@updatedAt

Artist

ARTIST PROFILES

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
UK slug	TEXT	URL key
name	TEXT	
born	INTEGER?	
died	INTEGER?	
country	TEXT?	
bio	TEXT?	
portrait	TEXT?	image URL
createdAt	TIMESTAMP(3)	default now()

Auction

A SALE / CATALOGUE

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
UK slug	TEXT	URL key
title	TEXT	
subtitle	TEXT?	
description	TEXT?	
currency	TEXT	default 'INR' (ISO 4217)
coverImage	TEXT?	
status	AuctionStatus	default UPCOMING
startsAt	TIMESTAMP(3)	bidding opens
endsAt	TIMESTAMP(3)	first lot closes
lotIntervalSeconds	INTEGER	default 60 · close stagger
createdAt / updatedAt	TIMESTAMP(3)	

INDEX (status, startsAt)

Enums

POSTGRES ENUM TYPES

Role — BIDDER · ADMIN
AuctionStatus — DRAFT → UPCOMING → LIVE → CLOSED
LotStatus — UPCOMING → LIVE → SOLD | PASSED · WITHDRAWN
ImageProvider — CLOUDINARY · EXTERNAL

Conventions

APPLIES TO ALL TABLES

IDs are **cuid** strings (collision-resistant, URL-safe).
Money is stored as **INTEGER** whole currency units — no floating point.
Timestamps are **TIMESTAMP(3)** (millisecond precision, UTC).
Schema is managed by **Prisma Migrate** (server/prisma/migrations).

3 · Database Schema (2/2)

Lot

ONE ARTWORK IN A SALE

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
FK auctionId	TEXT	→ Auction.id · ON DELETE CASCADE
lotNumber	INTEGER	1..n within auction
title	TEXT	
FK artistId	TEXT	→ Artist.id
year	TEXT?	
medium	TEXT?	
dimensions	TEXT?	
signature	TEXT?	
description	TEXT?	catalogue note
provenance	TEXT?	
estimateLow	INTEGER	whole currency units
estimateHigh	INTEGER	
startingBid	INTEGER	
reservePrice	INTEGER?	hidden from non-admins
currentBid	INTEGER?	denormalised live state
bidCount	INTEGER	default 0
FK highestBidderId	TEXT?	→ User.id
status	LotStatus	default UPCOMING
startsAt	TIMESTAMP(3)	
endsAt	TIMESTAMP(3)	moved by soft close
extensionCount	INTEGER	default 0
closedAt	TIMESTAMP(3)?	
version	INTEGER	optimistic lock counter
createdAt / updatedAt	TIMESTAMP(3)	

UNIQUE (auctionId, lotNumber) · INDEX (status, endsAt) · INDEX (status, startsAt)

Watchlist

SAVED LOTS

COLUMN	TYPE	CONSTRAINTS / NOTES
PK FK userId	TEXT	→ User.id · CASCADE
PK FK lotId	TEXT	→ Lot.id · CASCADE
createdAt	TIMESTAMP(3)	

PRIMARY KEY (userId, lotId)

LotImage

ARTWORK IMAGES

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
FK lotId	TEXT	→ Lot.id · CASCADE
provider	ImageProvider	CLOUDINARY | EXTERNAL
publicId	TEXT?	Cloudinary public_id
url	TEXT	secure URL
width	INTEGER?	px
height	INTEGER?	px
position	INTEGER	gallery order
alt	TEXT?	
createdAt	TIMESTAMP(3)	

INDEX (lotId, position)

Bid

APPEND-ONLY LEDGER

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
FK lotId	TEXT	→ Lot.id · CASCADE
FK userId	TEXT	→ User.id
amount	INTEGER	
isProxy	BOOLEAN	placed by max-bid engine
createdAt	TIMESTAMP(3)	ordered 1 ms apart per tx

UNIQUE (lotId, amount) · INDEX (lotId, createdAt) · INDEX (userId, createdAt)

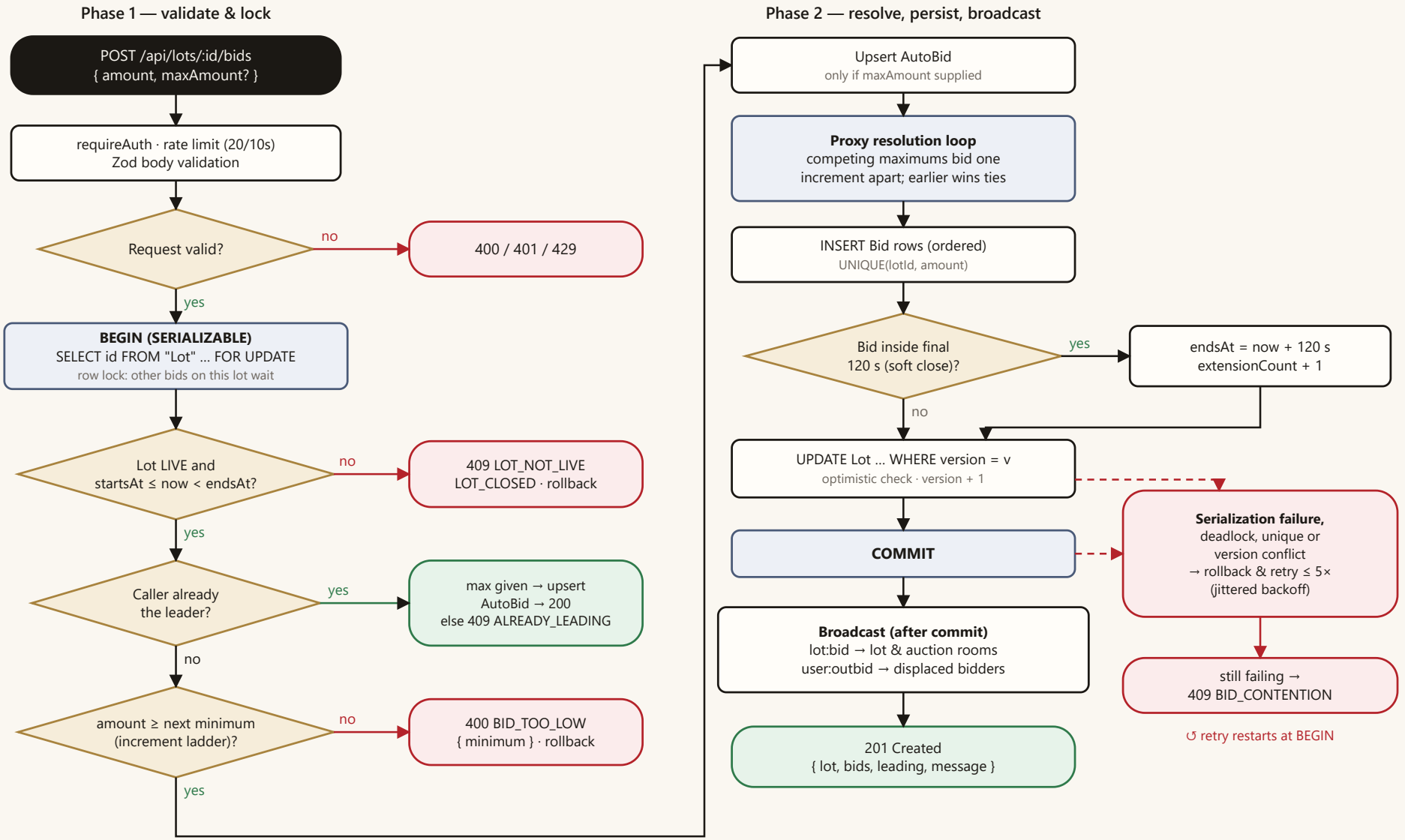
AutoBid

CONFIDENTIAL MAXIMUM

COLUMN	TYPE	CONSTRAINTS / NOTES
PK id	TEXT	cuid()
FK lotId	TEXT	→ Lot.id · CASCADE
FK userId	TEXT	→ User.id
maxAmount	INTEGER	ceiling
createdAt / updatedAt	TIMESTAMP(3)	earlier wins ties

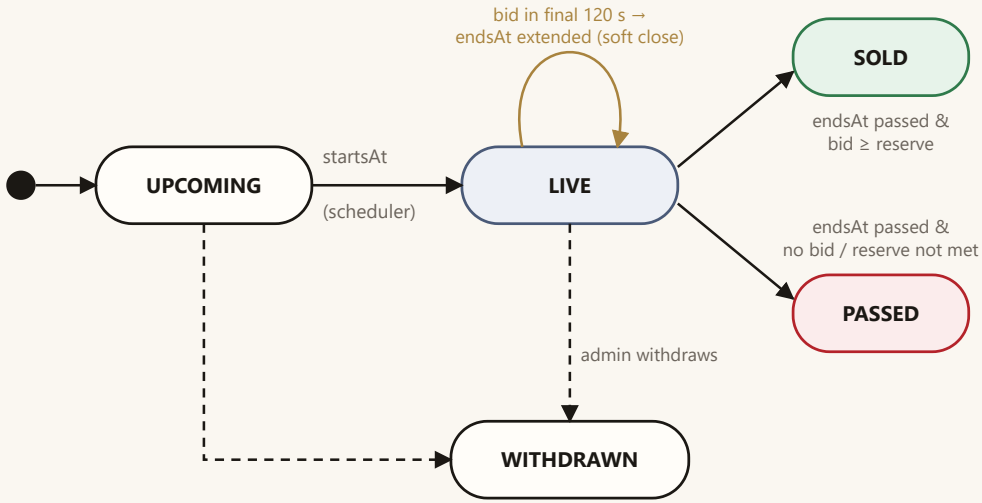
UNIQUE (lotId, userId) · INDEX (lotId, maxAmount)

4 · Flowchart — Bid Placement

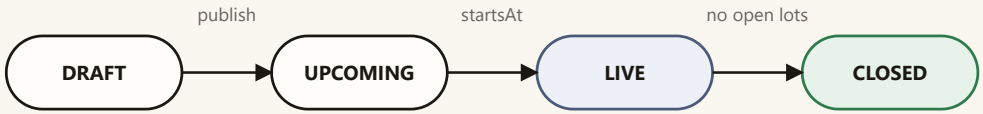


5 · Lifecycle & Scheduler

Lot state machine



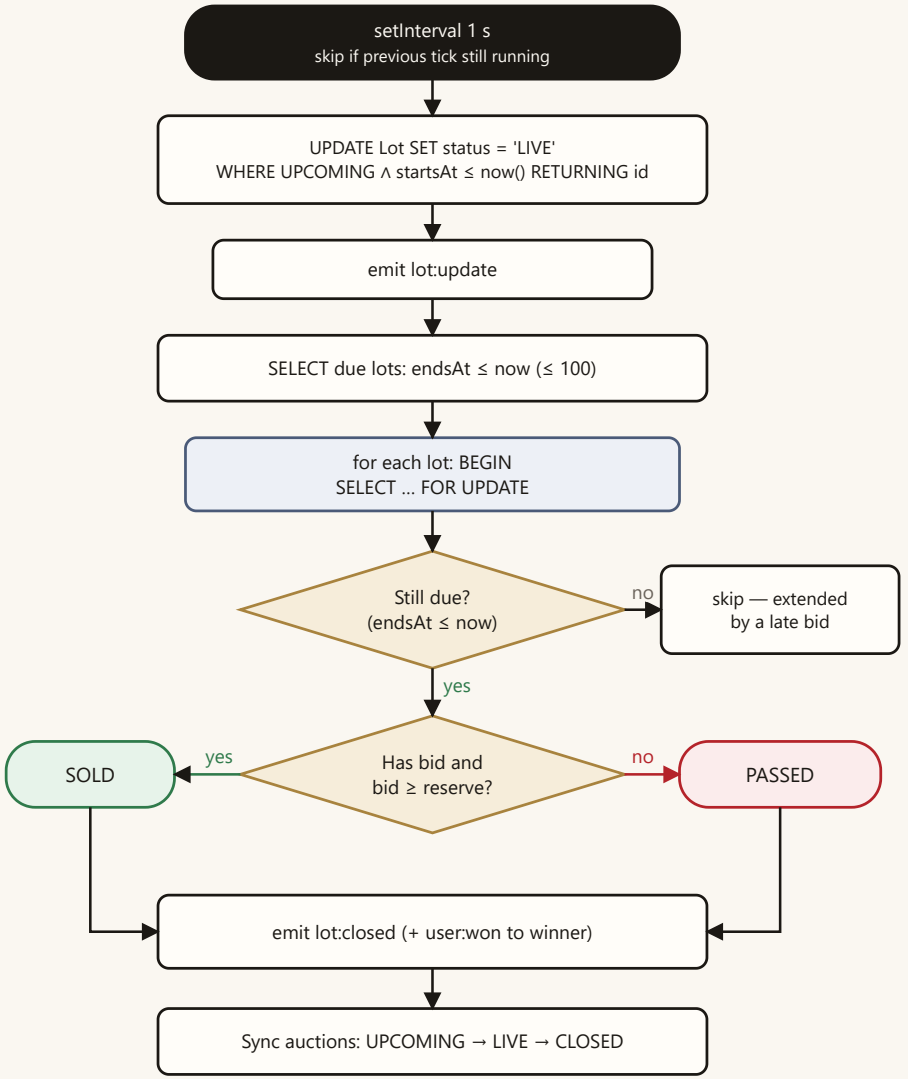
Auction state machine



Admins may create an auction directly as UPCOMING (publish = true).
Lots close in sequence: endsAt(n) = auction.endsAt + (n − 1) × lotIntervalSeconds.

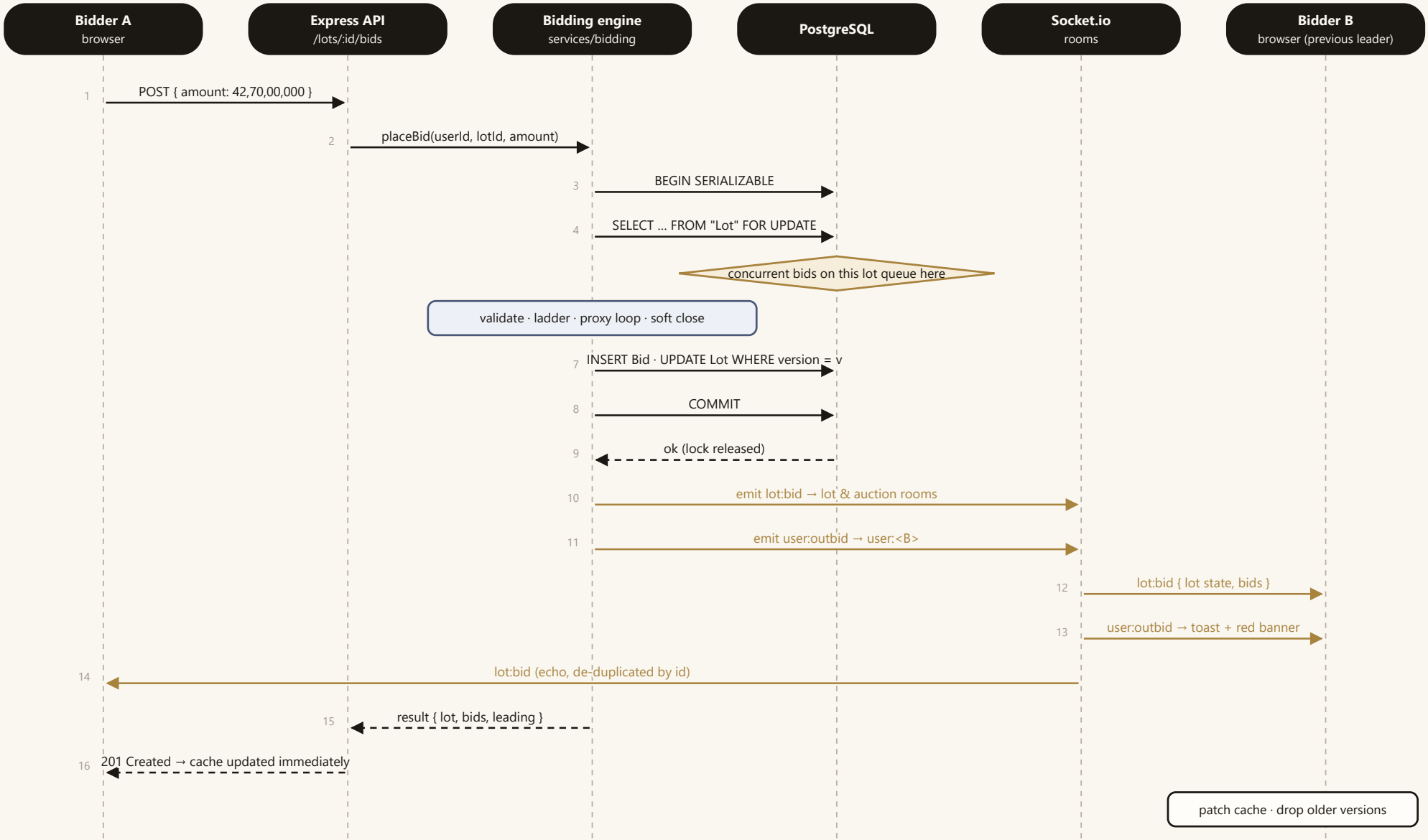
Why it is safe with several server instances
Opening uses one atomic UPDATE … RETURNING (one winner).
Closing re-checks endsAt under a row lock (FOR UPDATE),
so a lot extended by a last-second bid is never closed early.

Scheduler tick (every 1 s)



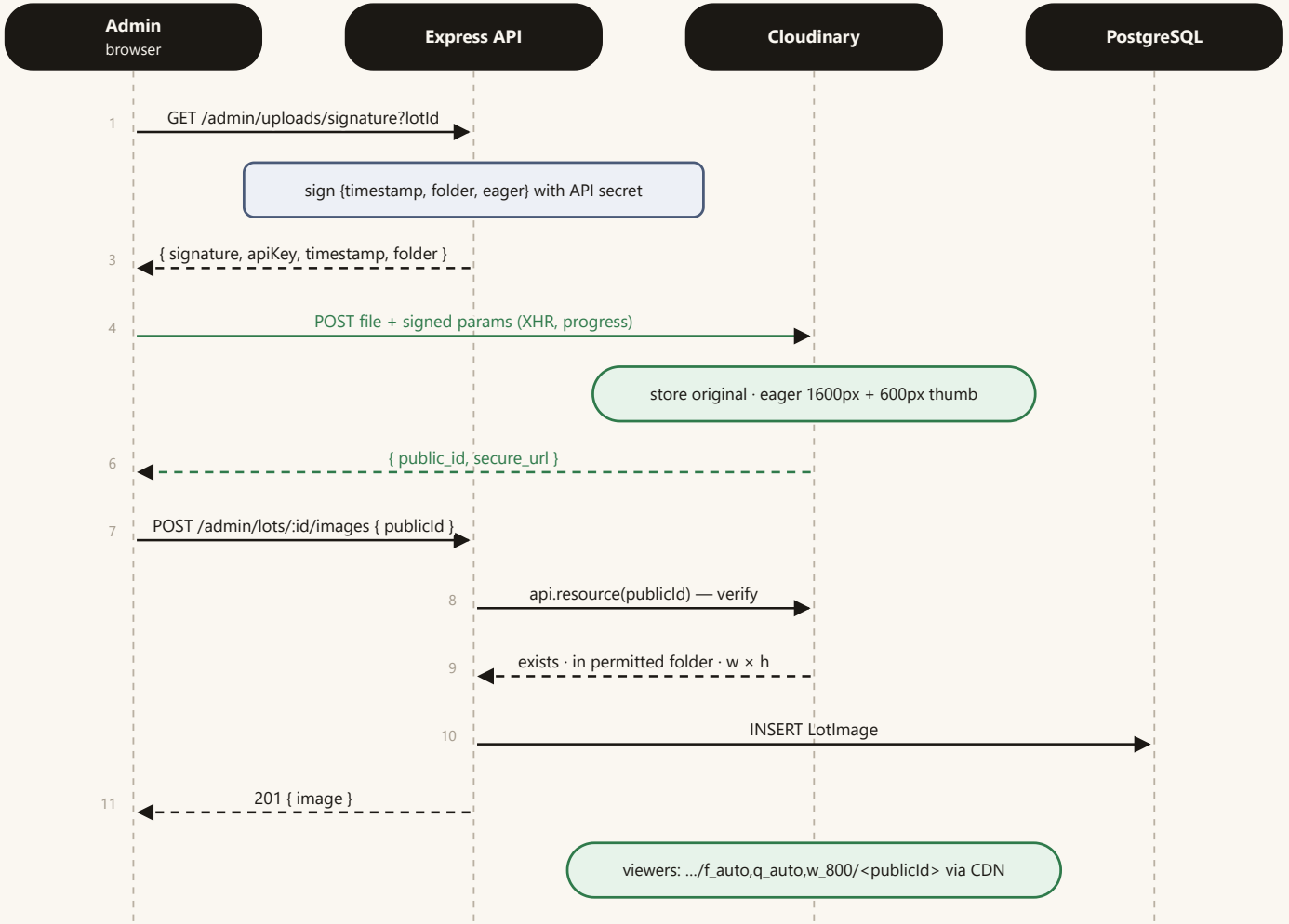
6 · Sequence — Live Bid Propagation

REST + WEBSOCKET

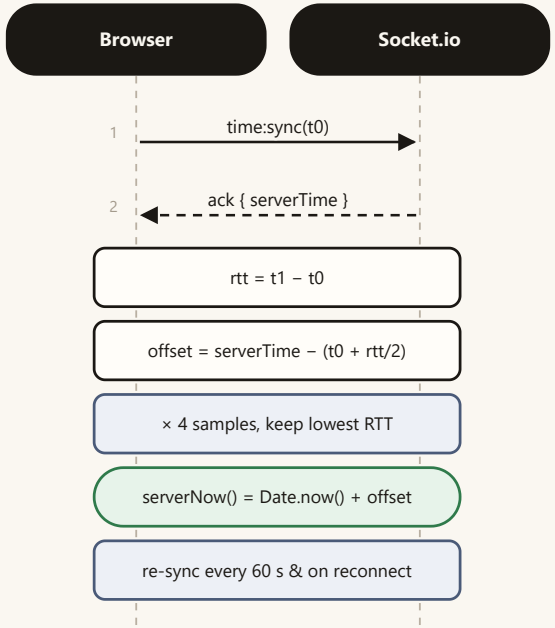


7 · Sequence — Image Upload & Clock Sync

Signed direct image upload (admin)



Server clock sync (countdowns)



Countdowns are cosmetic: the server rejects any bid at or after endsAt, whatever the client clock shows.

Deep zoom: shows the 1600 px rendition first; past 160 % zoom the full-resolution original is pre-loaded and swapped in seamlessly.